Folding hallmark	Vision — Δ top-1 accuracy	Language — Δ perplexity
C1 **Energy minimisation** *Thermodynamic folding funnel*	Langevin descent on $E_\theta(z \mid x)$ **+0.2 pp**	Energy-gated attention **+60.9 PPL** *(Langevin refinement: no effect)*
C2 **Adaptive topology** *Native contact formation*	Multi-head attention over particles (once at init) **+1.3 pp**	Standard per-layer self-attention *— subsumed by standard Transformer*
C3 **Cooperativity** *Local → global coupling*	Depthwise-separable 1D convolution **+0.5 pp**	Feed-forward network *— subsumed by standard Transformer*
C4 **Chaperone** *Stress-gated assisted folding*	$\lVert\nabla_z E\rVert$-gated corrective module **+0.4 pp**	Intra-block pre/post-FFN comparison **+4.9 PPL**
C5 **Environmental sensitivity** *Cellular-context modulation*	FiLM-style context scale & shift **+0.4 pp**	Layer normalisation *— subsumed by standard Transformer*

Contributions are the performance drop when each module is removed from the full model (Table: cross-domain decomposition). Vision: CIFAR-10Δ top-1 accuracy. Language: WikiText-2 Δ perplexity.